

DMT 14A - AIRFLOW & INJECTION

TOWARDS GREENER ENGINE OPERATIONS

1. PROJECT OVERVIEW

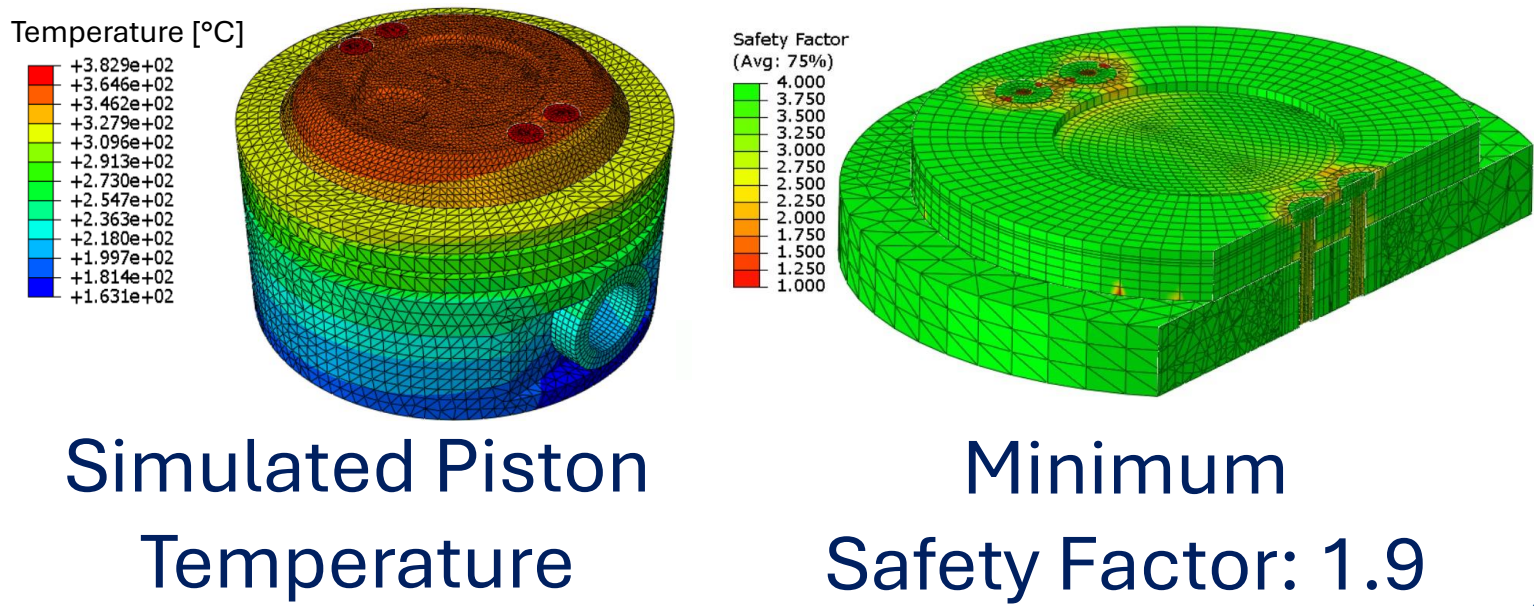
Compression ignition of petrol can be used to reduce emissions and improve engine efficiency. A test rig with modified commercial petrol engine was developed to operate in this regime with hydrogen peroxide (H₂O₂) as a pilot fuel. The system consists of *piston crown inserts*, *fuel delivery system* and *airflow system*.

2. PISTON CROWN INSERTS

OBJECTIVE: Increase compression ratio to promote auto-ignition.

DESIGN & FEATURES:

- Variable compression ratios of 8.6, 11 and 13 achieved
- Designed to withstand 2500K combustion temperatures
- Shape *optimised for in-cylinder flow and combustion*
- Secured by 4xM3 screws with preload and thread locker to mitigate vibrations

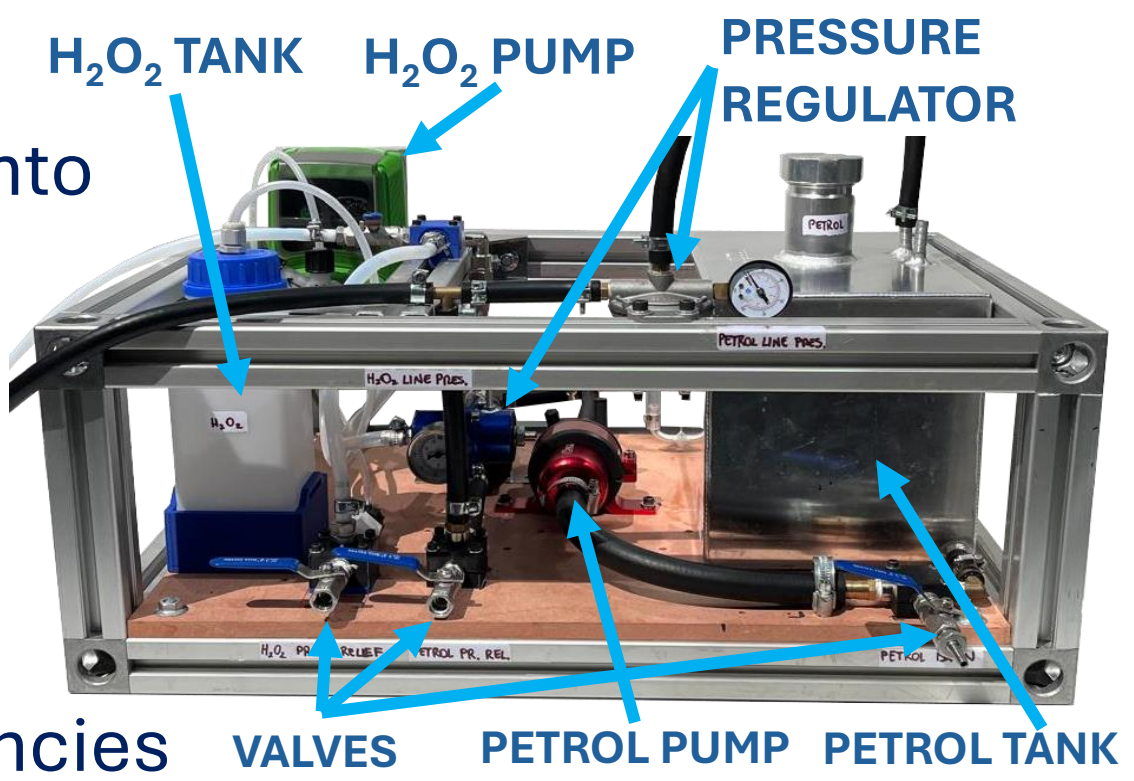


3. FUEL SYSTEM

OBJECTIVE: Deliver atomised petrol and hydrogen peroxide into the runner manifold.

DELIVERY SYSTEM DESIGN:

- Separate chassis for *safety* designed with team 14C
- H₂O₂ resistant pump
- Quick-disconnect for emergencies

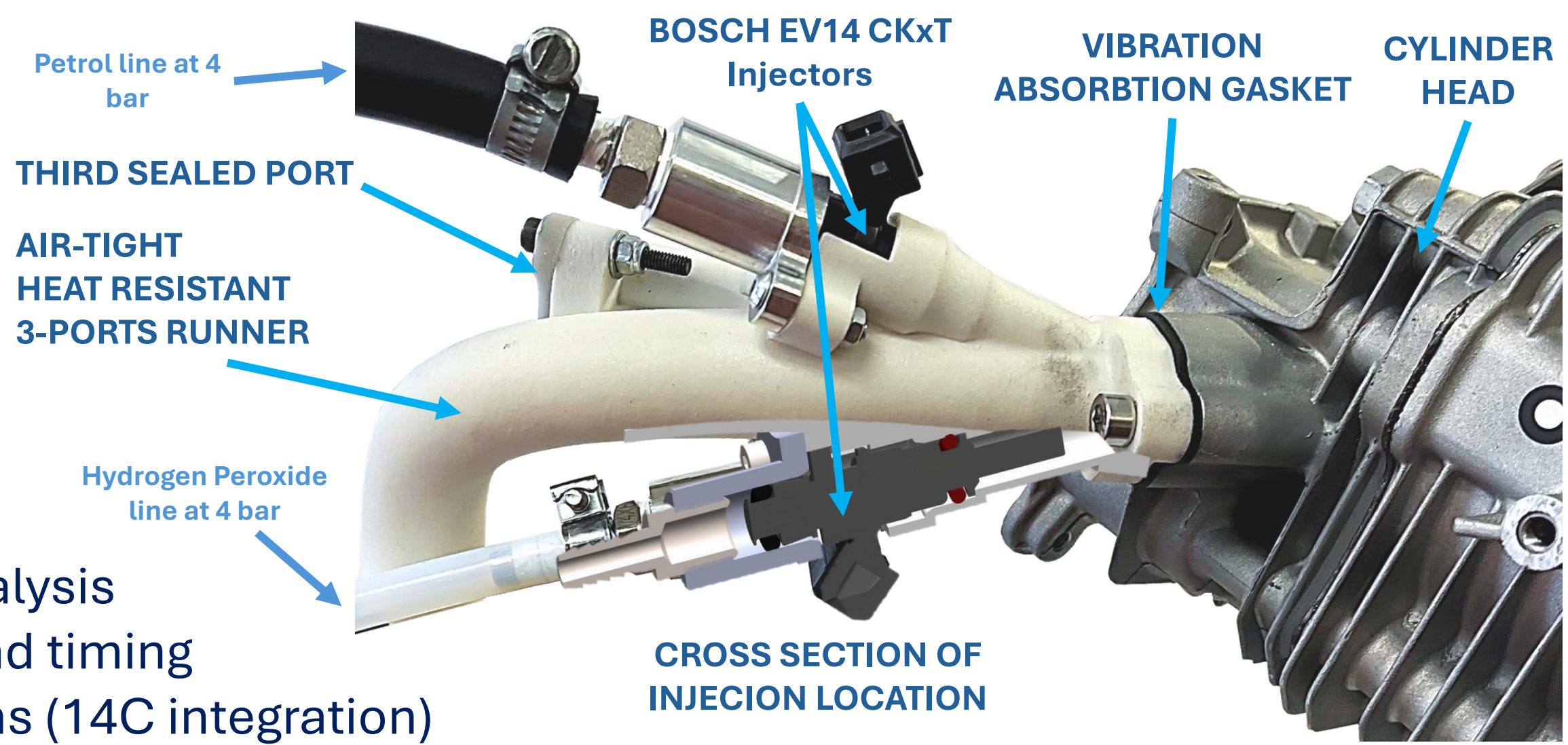


4. AIR FLOW SYSTEM

OBJECTIVE: Deliver well-mixed petrol and H₂O₂ air vapor mixture into the engine cylinder.

DESIGN & FEATURES:

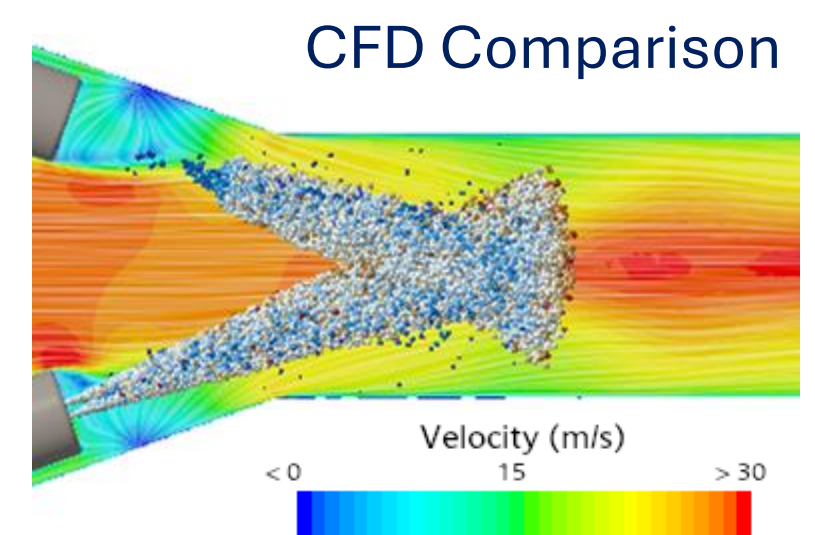
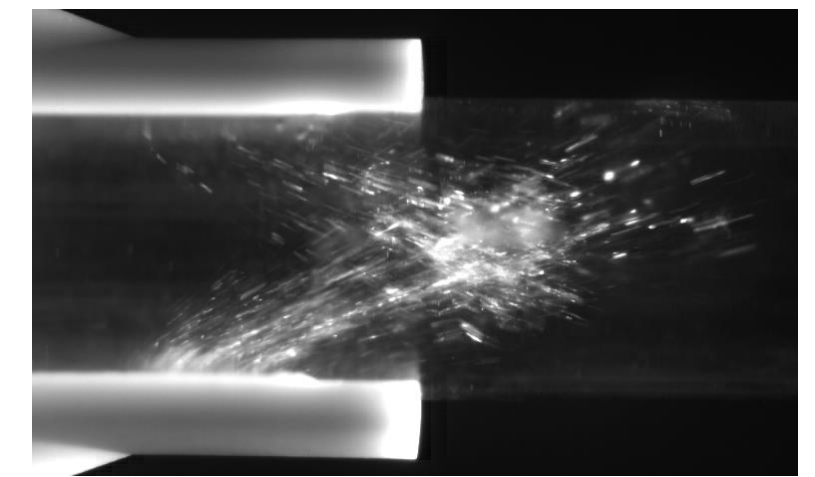
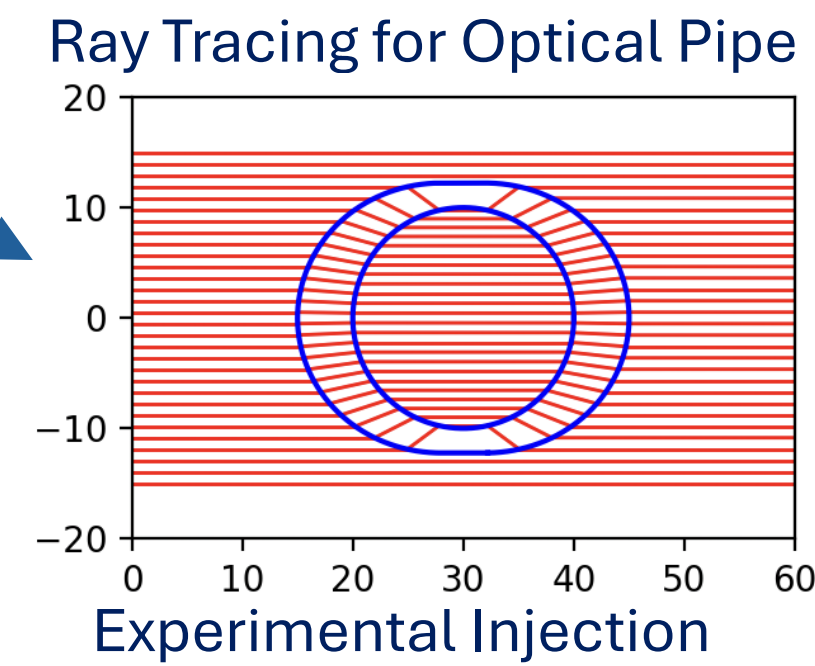
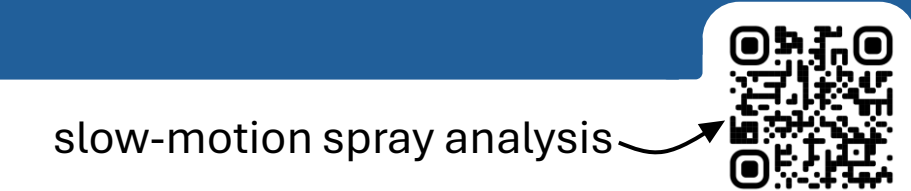
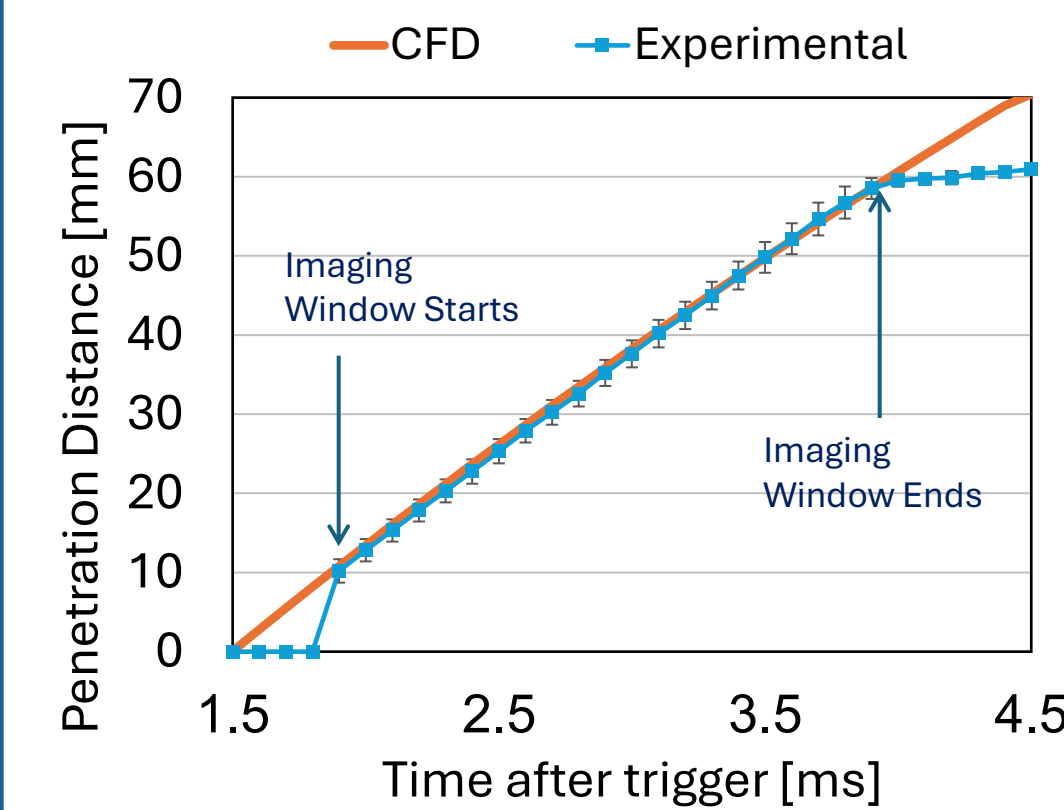
- Optimised injection *angle of 20°* from CFD analysis
- Fuel injectors used for precise fuel volumes and timing
- Bosch PST4 sensor to *monitor intake* conditions (14C integration)



5. TESTING PHASE

Flow Visualisation

- Custom pipe designed to remove optical distortion
- 10,000 FPS high-speed imaging used to visualise injection of sprays
- Experimental spray penetration matches CFD within ±2 mm



Thermal Stress Test

- Lumped capacitance model of the insert was confirmed with a safety margin against melting of 174 K
- Minimum yielding safety factor of 1.9 was confirmed

6. FUTURE WORK

Future work should focus on designing direct in-cylinder injection for precise combustion control and consider using 30% hydrogen peroxide concentrations.